HILBERT COEFFICIENTS OF COMPLETE BIPARTITE QUADRATIC IDEALS

KNOWLEDGE SENSE

Abstract. Let k be a field and let $1 \le p \le q$. In the polynomial ring S over k in variables $x_1, \dots, x_p, y_1, \dots, y_q$ consider the ideal $I_{p,q}$ generated by the squares of all variables and by all products $x_a y_b$, that is, the edge ideal of the complete bipartite graph $K_{p,q}$ together with all squares. We compute the lengths of $S/I_{p,q}^s$ for all $s \ge 1$ and all Hilbert coefficients of $I_{p,q}$. The Hilbert–Samuel polynomial gives $\ell(S/I_{p,q}^s)$ for every $s \ge 1$, the last coefficient e_{p+q} vanishes, and $e_0, \dots, e_p$ coincide with the Hilbert coefficients of the square of the maximal ideal. Each of the remaining coefficients $e_{p+1}, \dots, e_{p+q-1}$ is nonzero and has sign $(-1)^{p+1}$. Consequently all Hilbert coefficients are nonnegative if and only if the smaller part size p is odd. For $p = 2$ we give closed formulas, which exhibit $q - 1$ consecutive negative coefficients. The four-cycle $p = q = 2$, the smallest member of the family with a negative coefficient, has $e_3 = -2$ and is a counterexample to a conjecture of Fröberg on the nonnegativity of Hilbert coefficients of Artinian quadratic algebras. In Fröberg's list of predicted coefficients in four variables this ideal appears with $e_3 = 2$; we show that the sign is negative. The ideal is a fiber product, and the proof starts from the known decomposition of powers of fiber products.

1. Introduction

Let $S = k[z_1, \dots, z_n]$ be a polynomial ring over a field k, and let $I \subseteq S$ be a homogeneous ideal such that S/I is Artinian. By a theorem of Samuel [8] the length $\ell(S/I^s)$ is a polynomial in s of degree n for large s. Following Fröberg [3] we write it as

$$\ell(S/I^s) = \sum_{i=0}^{n} (-1)^i e_i(I) \binom{s+n-i-1}{n-i} \qquad (s \gg 0), \tag{1.1}$$

and call $e_0(I), \dots, e_n(I)$ the *Hilbert coefficients* of I. The same convention defines the Hilbert coefficients of an $\mathfrak{m}$-primary ideal of a Cohen–Macaulay local ring $(R, \mathfrak{m})$. For such ideals e_0 is positive, Northcott [7] showed that $e_1 \ge 0$, and Narita [5] showed that $e_2 \ge 0$ and gave an example with $e_3 < 0$; see also the introductions of [3] and [4].

The paper [3] studies Hilbert coefficients of Artinian quadratic algebras. On the basis of extensive calculations, Fröberg conjectured in this setting that negative coefficients do not occur [3, Conjecture 1.1]: if I is generated by quadrics and S/I is Artinian, then $e_i(I) \ge 0$ for all i. In the same paper he computed or predicted the Hilbert coefficients of all Artinian quotients by quadratic monomial ideals in at

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most four variables [3, Theorem 3.1, Conjecture 3.2], and conjectured a formula for the Hilbert coefficients of the square of the maximal ideal [3, Conjecture 3.1].

In this paper we study the following family. For integers $1 \le p \le q$ put $n = p+q$ and

$$S = k[x_1, \dots, x_p, y_1, \dots, y_q], \qquad I_{p,q} = \big(x_a^2,\ y_b^2,\ x_a y_b \ :\ 1 \le a \le p,\ 1 \le b \le q\big).$$

Thus $I_{p,q}$ is the edge ideal of the complete bipartite graph $K_{p,q}$ together with the squares of all variables; in the notation of [3] it is the ideal $I_{n,M}$ with M the edge set of $K_{p,q}$. Let $\mathfrak{m}_x = (x_1, \dots, x_p)$, $\mathfrak{m}_y = (y_1, \dots, y_q)$, $Q_x = (x_1^2, \dots, x_p^2)$ and $Q_y = (y_1^2, \dots, y_q^2)$. Then

$$I_{p,q} = Q_x + Q_y + \mathfrak{m}_x \mathfrak{m}_y$$

is the *fiber product* of $Q_x \subseteq k[x]$ and $Q_y \subseteq k[y]$ in the sense of Nguyen and Vu [6] and of Do, Nguyen and Seyed Fakhari [2]; correspondingly $S/I_{p,q} \cong A \times_k B$ with $A = k[x]/Q_x$ and $B = k[y]/Q_y$ [2, Section 1].

To state the result we use the following notation. For a power series f and an integer r, $[z^r]\, f$ denotes the coefficient of z^r in f, which is 0 if $r < 0$. For integers $a, b \ge 1$ and $s \ge 1$ let

$$D_{a,b}(s) = \sum_{\substack{t \ge 0,\ r \ge 1 \\ t+r \le s}} \ \sum_{\kappa = t+2r}^{a} \binom{t+b-1}{b-1} \binom{a}{\kappa} \binom{s-t-r+a-1}{a-1}, \tag{1.2}$$

where an empty inner sum is 0, and for integers $m \ge 0$ and $i \ge 1$ let

$$\Delta_{m,i}(z) = \sum_{j=m}^{i-1} (-1)^j \binom{i-1}{j} (1-z)^j, \tag{1.3}$$

which is 0 if $m \ge i$.

Theorem 1.1. *Let k be a field, let $1 \le p \le q$, $n = p+q$ and $I = I_{p,q}$.*

(a) *For every integer $s \ge 1$,*

$$\ell(S/I^s) = \binom{2s+n-1}{n} + D_{p,q}(s) + D_{q,p}(s).$$

(b) *Equation* (1.1) *holds for every integer $s \ge 1$, with $e_0(I) = 2^n$ and*

$$e_i(I) = [z^{n-i-1}]\, \frac{z^{i-1} - \Delta_{p,i}(z) - \Delta_{q,i}(z)}{(1-2z)^{i+1}} \qquad (1 \le i \le n).$$

In particular $e_i(I) = 2^{n-2i}\binom{n-i}{i} > 0$ for $0 \le i \le p$, and $e_n(I) = 0$.

(c) *For $p+1 \le i \le n-1$ one has $(-1)^{p+1} e_i(I) > 0$.*

(d) *All Hilbert coefficients of I are nonnegative if and only if p is odd. If p is even, then $e_{p+1}(I), \dots, e_{n-1}(I)$ are $q-1 \ge 1$ consecutive negative coefficients.*

For $p = q = 1$ the range in (c) is empty, and $I_{1,1}$ is the square of the maximal ideal of $k[x_1, y_1]$. For $p = 2$ the formula in (b) can be made explicit.

Corollary 1.2. *Let $q \ge 2$, $n = q+2$ and $I = I_{2,q}$. Then $e_0(I) = 2^n$, $e_1(I) = (n-1)2^{n-2}$, $e_2(I) = \binom{n-2}{2} 2^{n-4}$, $e_n(I) = 0$,*

$$e_{n-1}(I) = 3 - n + (-1)^{n-1},$$

and for $3 \le i \le n-2$, *with* $r = n-i-1 \ge 1$,

$$\begin{aligned} e_i(I) &= (2-i)\,2^r\binom{r+i}{i} + (i-1)\,2^{r-1}\binom{r+i-1}{i} \\ &= -\frac{2^{r-1}}{i}\binom{r+i-1}{i-1}\big(2i(i-2)+(i-3)r\big). \end{aligned}$$

In particular $e_3(I), \dots, e_{n-1}(I)$ *are* $n-3$ *consecutive negative coefficients.*

By Theorem 1.1(d), a negative coefficient requires p even, hence $p, q \ge 2$ and $n \ge 4$. The smallest member of the family with a negative Hilbert coefficient is therefore the four-cycle $C_4 = K_{2,2}$.

Corollary 1.3. *For* $p = q = 2$ *and every* $s \ge 1$,

$$\ell(S/I_{2,2}^s) = \binom{2s+3}{4} + 2s = 16\binom{s+3}{4} - 12\binom{s+2}{3} + \binom{s+1}{2} + 2\binom{s}{1},$$

so that $(e_0, e_1, e_2, e_3, e_4) = (16, 12, 1, -2, 0)$. *In particular* $I_{2,2}$ *is a counterexample to* [3, Conjecture 1.1].

The first lengths are 7, 39 and 132. Written in four variables, the ideal of Corollary 1.3 is the case $M = \{x_1x_2, x_2x_3, x_3x_4, x_4x_1\}$ of [3, Conjecture 3.2]. There Fröberg predicts the length formula above and lists the Hilbert coefficients as $(16, 12, 1, 2)$. Under the convention (1.1), which is the one of [3, p. 315], the predicted lengths give $e_3 = -2$, so the listed value of e_3 has the wrong sign. The predicted lengths themselves are correct by Theorem 1.1(a).

Relation to earlier work. The fiber-product structure of $I_{p,q}$ is the starting point of the proof. The powers of a fiber product $F = I + J + \mathfrak{mn}$ were decomposed by Nguyen and Vu [6, Proposition 4.4], and the resulting expansion $F^s = \sum_{j=0}^{s} (\mathfrak{mn})^{s-j}(I^j + J^j)$ was recorded by Do, Nguyen and Seyed Fakhari [2, Lemmas 2.10 and 5.6]; we use it in the form of Lemma 2.2. What is new here is the explicit evaluation of all lengths and of all Hilbert coefficients for the family $I_{p,q}$, and the sign classification.

Within Fröberg's tables, the family $I_{p,q}$ contains the square of the maximal ideal in two variables ($p = q = 1$), the case $M = \{x_1x_2, x_1x_3\}$ of [3, Theorem 3.1] ($p = 1$, $q = 2$), and the cases $M = \{x_1x_2, x_1x_3, x_1x_4\}$ and $M = \{x_1x_2, x_2x_3, x_3x_4, x_4x_1\}$ of [3, Conjecture 3.2] ($p = 1$, $q = 3$ and $p = q = 2$); see Remark 6.2. The formula for $e_0, \dots, e_p$ in Theorem 1.1(b) comes from the square of the maximal ideal. For this ideal Proposition 3.3 proves the coefficient formula of [3, Conjecture 3.1], together with the accompanying binomial identity after a correction of one index; see the discussion following Proposition 3.3.

Negative higher Hilbert coefficients, also in consecutive positions, are known for $\mathfrak{m}$-primary ideals of regular local rings; the examples of Mafi and Naderi [4, Examples 2.1 and 2.2] have cubic generators or generators of mixed degrees. Theorem 1.1 concerns quadratic ideals: for the family $I_{p,q}$ it determines the signs of all Hilbert coefficients, by the parity of one part size.

Banerjee, Goel and Verma [1, Theorem 3.3, Example 3.11] computed lengths attached to edge ideals of complete bipartite graphs, but for a different object; see Remark 6.3.

Dependencies. The decomposition of powers in Lemma 2.2 is due to [6, 2]; we include a short proof under the hypotheses used here. The rest of the proof is an exact count of standard monomials and a manipulation of formal power series with rational coefficients.

Organization. Section 2 fixes conventions and derives a membership criterion for powers of $I_{p,q}$ from the fiber-product decomposition. Section 3 counts standard monomials and proves Theorem 1.1(a) together with the formula for $e_0, \dots, e_p$. Section 4 proves the coefficient formula of Theorem 1.1(b), and Section 5 determines the signs. Section 6 treats the cases $p = 1$ and $p = 2$, compares with the literature and lists the coefficients for small p and q.

2. Preliminaries

2.1. **Lengths and Hilbert coefficients.** All binomial coefficients $\binom{N}{c}$ below have $N \geq 0$, and they vanish unless $0 \leq c \leq N$. For a power series f we write $[z^r]\, f$ as in the introduction. The following lemma connects the convention (1.1) with generating functions.

Lemma 2.1. *Let $n \geq 1$ and let $L\colon \{1, 2, \dots\} \to \mathbb{Q}$ be a function with*

$$\sum_{s \geq 1} L(s) z^s = \frac{z\, h(z)}{(1-z)^{n+1}}$$

for a polynomial $h \in \mathbb{Q}[z]$ of degree at most n. Write $h(1+u) = \sum_{i=0}^{n} e_i u^i$. Then

$$L(s) = \sum_{i=0}^{n} (-1)^i e_i \binom{s+n-i-1}{n-i} \qquad \text{for every } s \geq 1.$$

Proof. We have $h(z) = \sum_i e_i (z-1)^i = \sum_i (-1)^i e_i (1-z)^i$, hence $zh(z)/(1-z)^{n+1} = \sum_i (-1)^i e_i\, z/(1-z)^{n-i+1}$, and the coefficient of z^s in $z/(1-z)^{n-i+1}$ is $\binom{s-1+n-i}{n-i}$ for every $s \geq 1$. □

Since the polynomials $\binom{s+n-i-1}{n-i}$, $0 \leq i \leq n$, form a basis of the polynomials in s of degree at most n, the numbers e_i in Lemma 2.1 are uniquely determined by L. If $L(s) = \ell(S/I^s)$ for an ideal I as in (1.1), they are the Hilbert coefficients of I, and (1.1) holds for every $s \geq 1$.

For a monomial ideal $J \subseteq S$ with S/J Artinian, $\ell(S/J)$ is the number of monomials not in J, the *standard monomials* of J. Powers of monomial ideals are monomial ideals, and membership of a monomial in a monomial ideal is a combinatorial condition; hence the lengths in Theorem 1.1 do not depend on k. They also do not change if S is replaced by its localization at the homogeneous maximal ideal.

2.2. **Monomials.** From now on $1 \leq p \leq q$, $n = p + q$, $S = k[x_1, \dots, x_p, y_1, \dots, y_q]$ and $I = I_{p,q}$. We write $\mathbb{N} = \{0, 1, 2, \dots\}$. Every monomial of S is $x^\alpha y^\gamma$ with $\alpha \in \mathbb{N}^p$ and $\gamma \in \mathbb{N}^q$. For $\alpha \in \mathbb{N}^p$ we put

$$|\alpha| = \sum_a \alpha_a, \qquad \omega(\alpha) = \#\{a : \alpha_a \text{ odd}\}, \qquad \lfloor \alpha/2 \rfloor = \sum_a \lfloor \alpha_a/2 \rfloor = \frac{|\alpha| - \omega(\alpha)}{2},$$

and similarly for $\gamma \in \mathbb{N}^q$.

2.3. **Powers of fiber products.** Let R and T be polynomial rings over k of positive dimension, with homogeneous maximal ideals $\mathfrak{m}$ and $\mathfrak{n}$, and let $I' \subseteq \mathfrak{m}^2$ and $J' \subseteq \mathfrak{n}^2$ be homogeneous ideals. We identify ideals of R and of T with their extensions to $R \otimes_k T$, and we use the convention that the zeroth power of an ideal is the unit ideal. The fiber product of I' and J' is $F = I' + J' + \mathfrak{mn}$ [6, Notation 4.3].

Lemma 2.2 (Nguyen–Vu, Do–Nguyen–Seyed Fakhari)**.** *In this situation, for every $s \geq 1$,*

$$F^s = \sum_{j=0}^{s} (\mathfrak{mn})^{s-j} \big({I'}^j + {J'}^j \big).$$

Proof. Put $P = \mathfrak{mn}$. Since $F = I' + J' + P$, the ideal F^s is the sum of the ideals ${I'}^a {J'}^b P^c$ with $a + b + c = s$. If $a, b \geq 1$, then $I'J' \subseteq \mathfrak{m}^2\mathfrak{n}^2 = P^2$ gives ${I'}^a {J'}^b P^c \subseteq {I'}^{a-1} {J'}^{b-1} P^{c+2}$, and the exponents on the right again sum to s. Repeating this step, every summand is contained in one with $a = 0$ or $b = 0$, that is, in $P^{s-j} {I'}^j$ or $P^{s-j} {J'}^j$ for some $0 \leq j \leq s$. These ideals are themselves summands of F^s. $\square$

Lemma 2.2 is [2, Lemma 2.10], and it also follows from [6, Proposition 4.4(1)], namely $F^s = H^s + \sum_{i=1}^{s} (\mathfrak{mn})^{s-i} {J'}^i$ with $H = I' + \mathfrak{mn}$, combined with the binomial expansion of H^s. In arXiv:2409.17551v1, [2, Lemma 2.10] is stated for arbitrary proper homogeneous ideals, but its proof uses the inclusion $I'J' \subseteq \mathfrak{m}^2\mathfrak{n}^2$, which holds under the hypotheses above. We use the lemma only under these hypotheses. Some such hypothesis is needed: for $R = k[x]$, $T = k[y]$, $I' = (x)$ and $J' = (y)$ one has $F = (x, y)$, and $xy \in F^2$ does not lie in the right-hand side for $s = 2$.

We apply Lemma 2.2 with $R = k[x]$, $T = k[y]$, $I' = Q_x$ and $J' = Q_y$, so that $F = I_{p,q}$.

Lemma 2.3. *Let $s \geq 0$. A monomial $x^\alpha y^\gamma$ lies in $I_{p,q}^s$ if and only if*

$$|\alpha| + |\gamma| \geq 2s, \qquad \lfloor \alpha/2 \rfloor + |\gamma| \geq s, \qquad \lfloor \gamma/2 \rfloor + |\alpha| \geq s.$$

Proof. For $s = 0$ there is nothing to prove, so let $s \geq 1$. By Lemma 2.2, I^s is the sum of the monomial ideals $\mathfrak{m}_x^{s-j} Q_x^j \, \mathfrak{m}_y^{s-j}$ and $\mathfrak{m}_x^{s-j} \, \mathfrak{m}_y^{s-j} Q_y^j$, $0 \leq j \leq s$. A monomial lies in a sum of monomial ideals if and only if it lies in one of them. If $\mathfrak{a} \subseteq k[x]$ and $\mathfrak{b} \subseteq k[y]$ are monomial ideals, then $x^\alpha y^\gamma \in \mathfrak{ab}$ if and only if $x^\alpha \in \mathfrak{a}$ and $y^\gamma \in \mathfrak{b}$, because $\mathfrak{ab}$ is generated by products of a generator of $\mathfrak{a}$ and a generator of $\mathfrak{b}$. Moreover $y^\gamma \in \mathfrak{m}_y^{s-j}$ if and only if $|\gamma| \geq s - j$, and

$$x^\alpha \in \mathfrak{m}_x^{s-j} Q_x^j \iff \lfloor \alpha/2 \rfloor \geq j \text{ and } |\alpha| \geq s + j.$$

Indeed, $\mathfrak{m}_x^{s-j} Q_x^j$ is generated by the monomials $x^{2\beta+\delta}$ with $|\beta| = j$ and $|\delta| = s - j$. If such a monomial divides x^α, then $\beta_a \leq \lfloor \alpha_a/2 \rfloor$ for all a, and $|\alpha| \geq 2j + (s - j)$. Conversely, if $\lfloor \alpha/2 \rfloor \geq j$ and $|\alpha| \geq s + j$, choose β with $|\beta| = j$ and $\beta_a \leq \lfloor \alpha_a/2 \rfloor$; then $\alpha - 2\beta \in \mathbb{N}^p$ has degree $|\alpha| - 2j \geq s - j$ and contains some δ with $|\delta| = s - j$.

Hence $x^\alpha y^\gamma \in I^s$ if and only if one of the following holds for some $0 \leq j \leq s$:

(i) $\lfloor \alpha/2 \rfloor \geq j$, $|\alpha| \geq s + j$ and $|\gamma| \geq s - j$;
(ii) $\lfloor \gamma/2 \rfloor \geq j$, $|\gamma| \geq s + j$ and $|\alpha| \geq s - j$.

We claim that (i) holds for some j if and only if

$$|\alpha| \geq s, \qquad |\alpha| + |\gamma| \geq 2s, \qquad \lfloor \alpha/2 \rfloor + |\gamma| \geq s. \tag{2.1}$$

If (i) holds, then $|\alpha| \geq s + j \geq s$, $|\alpha| + |\gamma| \geq 2s$ and $\lfloor \alpha/2 \rfloor + |\gamma| \geq j + (s - j) = s$. Conversely, assume (2.1) and take $j = \max\{0, s - |\gamma|\} \leq s$. If $j = 0$, condition (i)

reduces to $|\alpha| \geq s$. If $j = s - |\gamma| > 0$, then $\lfloor \alpha/2 \rfloor \geq s - |\gamma| = j$ and $|\alpha| \geq 2s - |\gamma| = s + j$, and $|\gamma| = s - j$. This proves the claim, and symmetrically (ii) holds for some j if and only if $|\gamma| \geq s$, $|\alpha| + |\gamma| \geq 2s$ and $\lfloor \gamma/2 \rfloor + |\alpha| \geq s$.

If (2.1) holds, then $\lfloor \gamma/2 \rfloor + |\alpha| \geq |\alpha| \geq s$, so the three conditions of the lemma hold, and symmetrically for (ii). Conversely, if the three conditions hold, then $|\alpha| \geq s$ or $|\gamma| \geq s$, and accordingly (2.1) or its symmetric counterpart holds. □

Remark 2.4. Equivalently, the largest s with $x^\alpha y^\gamma \in I^s_{p,q}$ is

$$\min\Big\{ \Big\lfloor \frac{|\alpha| + |\gamma|}{2} \Big\rfloor,\ \lfloor \alpha/2 \rfloor + |\gamma|,\ \lfloor \gamma/2 \rfloor + |\alpha| \Big\}.$$

The first quantity is the bound coming from degrees. The second reflects the absence of the products $x_a x_{a'}$, $a \neq a'$, from $I_{p,q}$: in a product of generators dividing $x^\alpha y^\gamma$, at most $\lfloor \alpha/2 \rfloor$ factors are squares of x-variables, and every other factor involves a y-variable. The third is symmetric.

3. Lengths of powers

For integers $a, b \geq 1$ we let $D_{a,b}(s)$ be as in (1.2). Note that $D_{1,b} = 0$, because the inner sum requires $t + 2r \leq a$.

Proposition 3.1. *For every $s \geq 1$,*

$$\ell(S/I^s_{p,q}) = \binom{2s + n - 1}{n} + D_{p,q}(s) + D_{q,p}(s).$$

Proof. Write $I = I_{p,q}$. We count the standard monomials of I^s by means of Lemma 2.3. Every monomial of degree at most $2s - 1$ is standard, and there are $\binom{2s-1+n}{n}$ of them. A monomial $x^\alpha y^\gamma$ of degree at least $2s$ is standard if and only if $\lfloor \alpha/2 \rfloor + |\gamma| < s$ or $\lfloor \gamma/2 \rfloor + |\alpha| < s$. The two inequalities cannot hold simultaneously, because the sum of their left-hand sides is at least $|\alpha| + |\gamma| \geq 2s$.

We count the monomials of degree at least $2s$ with $\lfloor \alpha/2 \rfloor + |\gamma| < s$. Write $\alpha = 2\beta + \varepsilon$ with $\beta \in \mathbb{N}^p$ and $\varepsilon \in \{0,1\}^p$; this is a bijection between $\mathbb{N}^p$ and such pairs, and $|\varepsilon| = \omega(\alpha)$, $|\beta| = \lfloor \alpha/2 \rfloor$. Put $t = |\gamma|$, $\kappa = |\varepsilon|$, $l = |\beta|$ and $r = s - t - l$. The condition $\lfloor \alpha/2 \rfloor + |\gamma| < s$ says $r \geq 1$, and the degree condition $2l + \kappa + t \geq 2s = 2t + 2l + 2r$ says $\kappa \geq t + 2r$. Conversely, given $t \geq 0$, $r \geq 1$ with $t + r \leq s$ and $t + 2r \leq \kappa \leq p$, there are $\binom{t+q-1}{q-1}$ choices of γ, $\binom{p}{\kappa}$ choices of ε and $\binom{s-t-r+p-1}{p-1}$ choices of β with $|\beta| = s - t - r$. Hence these monomials are counted by $D_{p,q}(s)$. By symmetry the monomials of degree at least $2s$ with $\lfloor \gamma/2 \rfloor + |\alpha| < s$ are counted by $D_{q,p}(s)$. □

For $s = 1$ this gives $\ell(S/I_{p,q}) = 2^p + 2^q - 1$, the standard monomials being the squarefree monomials in the x-variables and those in the y-variables.

For integers $a, b \geq 1$ define the polynomial

$$P_{a,b}(z) = \sum_{\substack{t \geq 0,\ r \geq 1 \\ t + 2r \leq a}} \binom{t + b - 1}{b - 1} \Bigg(\sum_{\kappa = t + 2r}^{a} \binom{a}{\kappa} \Bigg) z^{t + r - 1}. \tag{3.1}$$

Thus $P_{1,b} = 0$, and for $a \geq 2$ the degree of $P_{a,b}$ is at most $a - 2$, because $t + r - 1 \leq t + 2r - 2 \leq a - 2$. For $a = 0$ the sum in (3.1) is empty, and we put $P_{0,b} = 0$.

Lemma 3.2. *For $a, b \geq 1$,*

$$\sum_{s\geq 1} D_{a,b}(s) z^s = \frac{z\, P_{a,b}(z)}{(1-z)^a}.$$

Proof. Substitute $l = s-t-r \geq 0$ in (1.2) and use $\sum_{l\geq 0} \binom{l+a-1}{a-1} z^l = (1-z)^{-a}$. □

The main term of Proposition 3.1 is the length of $S/\mathfrak{m}^{2s}$, where $\mathfrak{m} = \mathfrak{m}_x + \mathfrak{m}_y$. Its Hilbert coefficients are as follows.

Proposition 3.3. *Let $n \geq 1$ and $h_n(z) = \sum_{j\geq 0} \binom{n+1}{2j+1} z^j$. Then*

(i) $\displaystyle\sum_{s\geq 1} \binom{2s+n-1}{n} z^s = \frac{z\, h_n(z)}{(1-z)^{n+1}}$;

(ii) $\displaystyle h_n(1+u) = \sum_{i=0}^{\lfloor n/2\rfloor} 2^{n-2i} \binom{n-i}{i} u^i$.

Consequently, if $\mathfrak{m}$ is the homogeneous maximal ideal of a polynomial ring in n variables, then $e_i(\mathfrak{m}^2) = 2^{n-2i}\binom{n-i}{i}$ for $0 \leq i \leq \lfloor n/2\rfloor$, $e_i(\mathfrak{m}^2) = 0$ for $\lfloor n/2\rfloor < i \leq n$, and

$$\binom{2s+n-1}{n} = \sum_{i=0}^{\lfloor n/2\rfloor} (-1)^i 2^{n-2i} \binom{n-i}{i} \binom{s+n-i-1}{n-i} \qquad \text{for every } s \geq 1.$$

Proof. (i) From $\sum_{m\geq 0} \binom{m+n}{n} w^m = (1-w)^{-n-1}$ we obtain

$$\begin{aligned}\sum_{s\geq 1} \binom{2s-1+n}{n} w^{2s-1} &= \frac{(1-w)^{-n-1} - (1+w)^{-n-1}}{2} \\ &= \frac{(1+w)^{n+1} - (1-w)^{n+1}}{2(1-w^2)^{n+1}} \\ &= \frac{1}{(1-w^2)^{n+1}} \sum_{j\geq 0} \binom{n+1}{2j+1} w^{2j+1}.\end{aligned}$$

Multiplying by w and writing $z = w^2$ gives (i).

(ii) For $c \geq 1$ we have $\sum_{n\geq 0} \binom{n+1}{c} v^n = v^{c-1}/(1-v)^{c+1}$. Hence, in $\mathbb{Q}[u][[v]]$,

$$\begin{aligned}\sum_{n\geq 0} h_n(1+u) v^n &= \sum_{j\geq 0} \frac{(1+u)^j v^{2j}}{(1-v)^{2j+2}} = \frac{1}{(1-v)^2 - (1+u)v^2} \\ &= \frac{1}{1-v(2+uv)} = \sum_{m\geq 0} v^m (2+uv)^m,\end{aligned}$$

and the coefficient of $u^i v^n$ on the right is the coefficient of u^i in $(2+u)^{n-i}$, namely $\binom{n-i}{i} 2^{n-2i}$ if $2i \leq n$ and 0 otherwise.

Since $\ell(S/(\mathfrak{m}^2)^s) = \binom{2s+n-1}{n}$ and $\deg h_n = \lfloor n/2\rfloor \leq n$, the last assertions follow from Lemma 2.1. □

Proposition 3.3 proves the formula $e_j = \binom{n-j}{j} 2^{n-2j}$ conjectured in [3, Conjecture 3.1], including the assertion that $e_i(\mathfrak{m}^2) \neq 0$ exactly when $i \leq \lfloor n/2\rfloor$. The binomial identity stated there differs from the last display of Proposition 3.3 in its last binomial coefficient, which is printed as $\binom{s+n-j}{n-j}$; by (1.1) it should be

$\binom{s+n-j-1}{n-j}$. For $n = 2$ and $s = 1$ the printed right-hand side equals 10, whereas the left-hand side is 3. The numerical example [3, Example 3.1] is in the corrected form.

Corollary 3.4. *Let*

$$h_{p,q}(z) = h_n(z) + (1-z)^{q+1}P_{p,q}(z) + (1-z)^{p+1}P_{q,p}(z).$$

Then $\sum_{s\geq 1}\ell(S/I_{p,q}^s)z^s = z\,h_{p,q}(z)/(1-z)^{n+1}$, *and* $\deg h_{p,q} \leq n-1$. *Consequently* (1.1) *holds for every* $s \geq 1$, *with*

$$e_i(I_{p,q}) = [u^i]\,h_{p,q}(1+u) \qquad (0 \leq i \leq n).$$

Moreover $e_n(I_{p,q}) = 0$, $e_0(I_{p,q}) = 2^n$, *and* $e_i(I_{p,q}) = 2^{n-2i}\binom{n-i}{i} > 0$ *for* $0 \leq i \leq p$.

Proof. By Proposition 3.1, Lemma 3.2 and Proposition 3.3(i), the generating function of the lengths is

$$\frac{z\,h_n(z)}{(1-z)^{n+1}} + \frac{z\,P_{p,q}(z)}{(1-z)^p} + \frac{z\,P_{q,p}(z)}{(1-z)^q} = \frac{z\,h_{p,q}(z)}{(1-z)^{n+1}}.$$

Since $n \geq 2$, $\deg h_n = \lfloor n/2 \rfloor \leq n-1$, and $\deg (1-z)^{q+1}P_{p,q} \leq q+1+p-2 = n-1$; the same bound holds for the third summand. Lemma 2.1 gives (1.1) for all $s \geq 1$, the formula for e_i, and $e_n = 0$. At $z = 1+u$ the second and third summands of $h_{p,q}$ become $(-u)^{q+1}P_{p,q}(1+u)$ and $(-u)^{p+1}P_{q,p}(1+u)$, which are divisible by u^{p+1}. Hence $e_i(I_{p,q}) = e_i(\mathfrak{m}^2)$ for $i \leq p$, and Proposition 3.3 applies. Finally $\binom{n-i}{i} > 0$ for $2i \leq 2p \leq n$. □

This proves Theorem 1.1(a) and the statements of Theorem 1.1(b) that follow the words "in particular".

4. The coefficient formula

By Corollary 3.4 it remains to compute $[u^i]\,(-u)^{a+1}P_{b,a}(1+u)$ for $(a,b) = (q,p)$ and $(a,b) = (p,q)$. We first sum the polynomials $P_{b,a}$ over b.

Lemma 4.1. *Let* $a \geq 1$. *In* $\mathbb{Q}[z][[y]]$,

$$\sum_{b\geq 0} P_{b,a}(z)\,y^b = \frac{y^2(1-y)^a}{(1-2y)\big(1-(1+z)y\big)^a\big((1-y)^2 - zy^2\big)}.$$

Proof. For $\kappa \geq 0$ let $Q_\kappa(z) = \sum_{t\geq 0,\, r\geq 1,\, t+2r\leq\kappa}\binom{t+a-1}{a-1}z^{t+r-1}$; in particular $Q_0 = 0$. Exchanging the summations over κ and over (t,r) in (3.1) gives $P_{b,a} = \sum_{\kappa=0}^{b}\binom{b}{\kappa}Q_\kappa$ for all $b \geq 0$; for $b = 0$ both sides vanish, because $P_{0,a} = 0$ and $Q_0 = 0$. Summing over $t \geq 0$, $r \geq 1$ and $\kappa \geq t+2r$ independently,

$$\sum_{\kappa\geq 0} Q_\kappa(z)w^\kappa = \frac{1}{1-w}\cdot\frac{1}{(1-zw)^a}\cdot\frac{w^2}{1-zw^2}.$$

Since $\sum_{b\geq\kappa}\binom{b}{\kappa}y^b = y^\kappa/(1-y)^{\kappa+1}$, we get

$$\sum_{b\geq 0} P_{b,a}(z)\,y^b = \sum_{\kappa\geq 0} Q_\kappa(z)\frac{y^\kappa}{(1-y)^{\kappa+1}} = \frac{1}{1-y}\sum_{\kappa\geq 0} Q_\kappa(z)w^\kappa\Big|_{w=y/(1-y)}.$$

With $w = y/(1-y)$ we have $1-w = (1-2y)/(1-y)$, $1-zw = (1-(1+z)y)/(1-y)$ and $1-zw^2 = ((1-y)^2 - zy^2)/(1-y)^2$, and the stated formula follows. □

Proposition 4.2. *Let $a, b \geq 1$ and $i \geq 1$. Then*

$$[u^i]\left((-u)^{a+1}P_{b,a}(1+u)\right) = -[z^{a+b-i-1}]\,\frac{\Delta_{a,i}(z)}{(1-2z)^{i+1}}.$$

Proof. If $i \leq a$, both sides vanish: the left-hand side because of the factor u^{a+1}, the right-hand side because $\Delta_{a,i} = 0$. Let $i \geq a+1$. After multiplication by $(-1)^{a+1}$, the claim reads

$$[u^{i-a-1}]\,P_{b,a}(1+u) = (-1)^a\,[z^{a+b-i-1}]\,\frac{\Delta_{a,i}(z)}{(1-2z)^{i+1}}. \tag{4.1}$$

We prove (4.1) for all $b \geq 0$, where $P_{0,a} = 0$. The right-hand side vanishes for $b \leq i-a$, and its generating function over $b \geq 0$ is $(-1)^a y^{i+1-a}\Delta_{a,i}(y)/(1-2y)^{i+1}$. By Lemma 4.1 with $z = 1+u$, the generating function of the left-hand side over $b \geq 0$ is $[u^{i-a-1}]\,\Phi$, where

$$\Phi = \Phi(u,y) = \frac{y^2(1-y)^a}{(1-2y)\left(1-(2+u)y\right)^a(1-2y-uy^2)} \in \mathbb{Q}[u][[y]].$$

Comparing coefficients of the powers of u, we see that (4.1) holds for all $b \geq 0$ and all $i \geq a+1$ if and only if

$$\sum_{i\geq a+1} u^{i-a-1}\,\frac{y^{i+1-a}\,\Delta_{a,i}(y)}{(1-2y)^{i+1}} = (-1)^a\,\Phi \tag{4.2}$$

in $\mathbb{Q}[u][[y]]$; the series on the left converges y-adically, since its i-th term is divisible by y^{i+1-a}. Put $w = uy/(1-2y) \in y\mathbb{Q}[u][[y]]$. The i-th term on the left of (4.2) equals $y^2(1-2y)^{-a-2}\,w^{i-a-1}\Delta_{a,i}(y)$, and with $m = i-1$,

$$\begin{aligned}
\sum_{i\geq a+1} w^{i-a-1}\Delta_{a,i}(y) &= \sum_{j\geq a}(-1)^j(1-y)^j \sum_{m\geq j}\binom{m}{j}w^{m-a} \\
&= \sum_{j\geq a}(-1)^j(1-y)^j\,\frac{w^{j-a}}{(1-w)^{j+1}} \\
&= \frac{(-1)^a(1-y)^a}{(1-w)^{a+1}}\cdot\frac{1}{1+(1-y)w/(1-w)} \\
&= \frac{(-1)^a(1-y)^a}{(1-w)^a(1-yw)}.
\end{aligned}$$

Finally $(1-2y)^a(1-w)^a = (1-(2+u)y)^a$ and $(1-2y)(1-yw) = 1-2y-uy^2$, so that

$$\frac{y^2}{(1-2y)^{a+2}}\cdot\frac{(-1)^a(1-y)^a}{(1-w)^a(1-yw)} = (-1)^a\Phi.$$

This proves (4.2) and the proposition. □

Proof of Theorem 1.1(b). By Corollary 3.4 it remains to prove the formula for e_i, $1 \leq i \leq n$. By Corollary 3.4 and Proposition 3.3(ii),

$$e_i(I_{p,q}) = 2^{n-2i}\binom{n-i}{i} + [u^i]\,(-u)^{q+1}P_{p,q}(1+u) + [u^i]\,(-u)^{p+1}P_{q,p}(1+u).$$

The first term equals $[z^{n-2i}]\,(1-2z)^{-i-1} = [z^{n-i-1}]\,z^{i-1}/(1-2z)^{i+1}$, and both sides vanish if $2i > n$. By Proposition 4.2 with $(a,b) = (q,p)$ and $(a,b) = (p,q)$, the

second and third terms are $-[z^{n-i-1}]\,\Delta_{q,i}(z)/(1-2z)^{i+1}$ and $-[z^{n-i-1}]\,\Delta_{p,i}(z)/(1-2z)^{i+1}$. □

5. Signs

For $i \geq 2$ and $1 \leq m \leq i-1$ put

$$G_{m,i}(z) = z^{i-1} - \Delta_{m,i}(z) = \sum_{j=0}^{m-1} (-1)^j \binom{i-1}{j} (1-z)^j,$$

where the second expression follows from (1.3) and the binomial expansion $z^{i-1} = \big(1-(1-z)\big)^{i-1} = \sum_{j=0}^{i-1}(-1)^j\binom{i-1}{j}(1-z)^j$. For $r \geq 0$ put

$$\tau_{m,i}(r) = [z^r]\,\frac{G_{m,i}(z)}{(1-2z)^{i+1}}.$$

Lemma 5.1. *Let $p+1 \leq i \leq n-1$ and $r = n-i-1 \geq 0$. Then*

$$e_i(I_{p,q}) = \begin{cases} \tau_{p,i}(r) & \text{if } i \leq q, \\ \tau_{p,i}(r) + \tau_{q,i}(r) & \text{if } i > q. \end{cases}$$

Proof. Since $p < i$, the numerator in Theorem 1.1(b) is $G_{p,i} - \Delta_{q,i}$. If $i \leq q$ then $\Delta_{q,i} = 0$. If $i > q$, the numerator is $G_{p,i} + G_{q,i} - z^{i-1}$; since $n = p+q \leq 2i-2$, we have $r \leq i-3$, so z^{i-1} does not contribute to the coefficient of z^r. □

Lemma 5.2. *For integers $M \geq 0$ and $N \geq M+1$ put*

$$\beta_M(N) = (-1)^M \sum_{l=0}^{M} \Big(-\frac{1}{2}\Big)^l \binom{N}{l}.$$

Then $\beta_M(N) \geq 0$, and $\beta_0(N) = 1$.

Proof. The value $\beta_0(N) = 1$ is clear. We prove $\beta_M(N) \geq 0$ by induction on $N \geq 1$, simultaneously for all M with $0 \leq M \leq N-1$. The case $M = 0$ is clear. For $N = M+1$,

$$\sum_{l=0}^{M} \Big(-\frac{1}{2}\Big)^l \binom{M+1}{l} = \Big(1-\frac{1}{2}\Big)^{M+1} - \Big(-\frac{1}{2}\Big)^{M+1},$$

so $\beta_M(M+1) = 2^{-M-1}\big(1+(-1)^M\big) \geq 0$. This settles $N = 1$. For $1 \leq M \leq N-2$, Pascal's rule gives

$$\beta_M(N) = \beta_M(N-1) + \tfrac{1}{2}\,\beta_{M-1}(N-1),$$

and both terms on the right are nonnegative by the induction hypothesis for $N-1$, since $M \leq (N-1)-1$ and $M-1 \leq (N-1)-1$. □

Lemma 5.3. *Let $i \geq 2$ and $1 \leq m \leq i-1$. Then*

$$(-1)^{m-1} G_{m,i}(z) = \sum_{c=0}^{m-1} \gamma_c\,(1-2z)^c$$

with rational numbers $\gamma_c \geq 0$ and $\gamma_{m-1} > 0$. Consequently $(-1)^{m-1}\tau_{m,i}(r) > 0$ for every $r \geq 0$.

Proof. Write $v = 1-2z$, so that $1-z = (1+v)/2$. The coefficient of v^c in $G_{m,i}$ is

$$\sum_{j=c}^{m-1} (-1)^j 2^{-j} \binom{i-1}{j}\binom{j}{c} = (-1)^c 2^{-c} \binom{i-1}{c} \sum_{l=0}^{m-1-c} \Big(-\frac{1}{2}\Big)^l \binom{i-1-c}{l},$$

using $\binom{i-1}{j}\binom{j}{c} = \binom{i-1}{c}\binom{i-1-c}{j-c}$ and $l = j-c$. Put $M = m-1-c$ and $N = i-1-c \geq M+1$. Since $(-1)^c(-1)^M = (-1)^{m-1}$, the coefficient of v^c in $(-1)^{m-1}G_{m,i}$ is $\gamma_c = 2^{-c}\binom{i-1}{c}\beta_M(N)$, which is nonnegative by Lemma 5.2, and $\gamma_{m-1} = 2^{1-m}\binom{i-1}{m-1} > 0$. Hence

$$(-1)^{m-1}\tau_{m,i}(r) = \sum_{c=0}^{m-1} \gamma_c\, [z^r]\,(1-2z)^{c-i-1} = \sum_{c=0}^{m-1} \gamma_c\, 2^r \binom{r+i-c}{r} > 0,$$

since $c-i-1 < 0$ for all $c \leq m-1$. □

Lemmas 5.1 and 5.3 already give the sign of $e_i(I_{p,q})$ when $i \leq q$, and when $i > q$ and $p \equiv q \pmod 2$. For $i > q$ and $p \not\equiv q \pmod 2$ the two contributions have opposite signs, and we show that the contribution of the smaller part dominates. We use the following integral representation; here $\int_0^1$ denotes the $\mathbb{Q}$-linear map on polynomials in θ that sends θ^c to $1/(c+1)$, applied coefficientwise.

Lemma 5.4. *Let $i \geq 2$ and $1 \leq m \leq i-1$. In $\mathbb{Q}[z]$,*

$$G_{m,i}(z) - z^{i-1} = (-1)^{m-1}(i-1)\binom{i-2}{m-1}(1-z)^m \int_0^1 \theta^{m-1}\big(z+(1-z)\theta\big)^{i-m-1}\,d\theta.$$

Proof. For integers $N \geq m \geq 1$ we have in $\mathbb{Q}[X,Y]$

$$\sum_{j=m}^{N} \binom{N}{j} X^j Y^{N-j} = N\binom{N-1}{m-1} X^m \int_0^1 \theta^{m-1}\big(Y + X(1-\theta)\big)^{N-m}\,d\theta. \tag{5.1}$$

Indeed, by $\int_0^1 \theta^{m-1}(1-\theta)^l\,d\theta = (m-1)!\,l!/(m+l)!$ the coefficient of $X^{m+l}Y^{N-m-l}$ on the right is $N\binom{N-1}{m-1}\binom{N-m}{l}\frac{(m-1)!\,l!}{(m+l)!} = \binom{N}{m+l}$. Now $G_{m,i}(z) - z^{i-1} = -\Delta_{m,i}(z) = -\sum_{j=m}^{i-1}\binom{i-1}{j}(z-1)^j$. Apply (5.1) with $N = i-1$, $X = z-1$ and $Y = 1$, and note $Y + X(1-\theta) = z + (1-z)\theta$ and $X^m = (-1)^m(1-z)^m$. □

Lemma 5.5. *Let $f \in \mathbb{Q}[[z]]$ and $r \geq 0$. Then $[z^r]\,f(z) = [w^r]\,(1+2w)^{r-1} f\big(w/(1+2w)\big)$.*

Proof. Both sides depend only on the coefficients of $z^0, \dots, z^r$ in f and are linear in f, so it suffices to take $f = z^c$ with $0 \leq c \leq r$. The right-hand side is then $[w^{r-c}]\,(1+2w)^{r-1-c}$, which is 1 if $c = r$ and 0 if $c < r$, since then $(1+2w)^{r-1-c}$ is a polynomial of degree $r-1-c$. □

Proposition 5.6. *Let $i \geq 2$ and $0 \leq r \leq i-2$. For $1 \leq m \leq i-1$ put*

$$\upsilon_m = \frac{(-1)^{m-1}\tau_{m,i}(r)}{\binom{i-2}{m-1}}.$$

Then $\upsilon_1 \geq \upsilon_2 \geq \cdots \geq \upsilon_{i-1} > 0$.

Proof. Positivity is Lemma 5.3. Since $r < i-1$, the coefficient of z^r in $z^{i-1}/(1-2z)^{i+1}$ vanishes, so by Lemma 5.4

$$\upsilon_m = (i-1)\,[z^r]\,\frac{(1-z)^m}{(1-2z)^{i+1}}\int_0^1 \theta^{m-1}\big(z+(1-z)\theta\big)^{i-m-1}\,d\theta.$$

We apply Lemma 5.5. For $z = w/(1+2w)$ we have $1-2z = (1+2w)^{-1}$, $1-z = (1+w)/(1+2w)$ and $z+(1-z)\theta = (\theta+(1+\theta)w)/(1+2w)$. Since evaluation at $z = w/(1+2w)$ commutes with $\int_0^1$, and since $(r-1)+(i+1)-m-(i-m-1) = r+1$, we obtain

$$\upsilon_m = (i-1)\,[w^r]\,(1+2w)^{r+1}\int_0^1 F_m(\theta,w)\,d\theta,$$

where $F_m = (1+w)^m\theta^{m-1}\big(\theta+(1+\theta)w\big)^{i-m-1}$. For $1 \le m \le i-2$,

$$\begin{aligned} F_m - F_{m+1} &= (1+w)^m\theta^{m-1}\big(\theta+(1+\theta)w\big)^{i-m-2}\big(\theta+(1+\theta)w-(1+w)\theta\big)\\ &= w\,(1+w)^m\theta^{m-1}\big(\theta+(1+\theta)w\big)^{i-m-2}, \end{aligned}$$

a polynomial in θ and w with nonnegative coefficients. Multiplication by $(1+2w)^{r+1}$, the map $\int_0^1$ and the extraction of a coefficient preserve nonnegativity, so $\upsilon_m - \upsilon_{m+1} \ge 0$. □

Proof of Theorem 1.1(c) *and* (d). Let $p+1 \le i \le n-1$ and $r = n-i-1 \ge 0$. By Lemma 5.3, $\tau_{p,i}(r)$ has sign $(-1)^{p-1} = (-1)^{p+1}$, and if $q < i$ then $\tau_{q,i}(r)$ has sign $(-1)^{q+1}$. By Lemma 5.1, this proves (c) if $i \le q$, and also if $i > q$ and $p \equiv q \pmod 2$.

Let $i > q$ and $p \not\equiv q \pmod 2$, so $p < q < i$. As in Lemma 5.1, $r \le i-3$, so Proposition 5.6 applies with $m \in \{p,q\}$ and gives $\upsilon_p \ge \upsilon_q > 0$. We claim $\binom{i-2}{p-1} > \binom{i-2}{q-1}$. Put $N = i-2$ and $c = N-(q-1) \ge 0$. Then $\binom{N}{q-1} = \binom{N}{c}$, and $c < p-1 < N-c$, because $(p-1)+(q-1) = n-2 > N$ and $p < q$. Since $\min\{p-1, N-p+1\} > c$ and the binomial coefficients $\binom{N}{j}$ increase strictly for $0 \le j \le \lfloor N/2 \rfloor$, the claim follows. Hence

$$(-1)^{p+1}\tau_{p,i}(r) = \binom{i-2}{p-1}\upsilon_p > \binom{i-2}{q-1}\upsilon_q = (-1)^{q+1}\tau_{q,i}(r) > 0,$$

and $e_i(I_{p,q}) = \tau_{p,i}(r) + \tau_{q,i}(r)$ has the sign $(-1)^{p+1}$. This proves (c).

For (d), if p is odd then $e_0, \dots, e_{n-1}$ are positive by (b) and (c), and $e_n = 0$. If p is even, then $p \ge 2$, so $q \ge 2$ and $n-1 \ge p+1$; by (c) the $q-1$ coefficients $e_{p+1}, \dots, e_{n-1}$ are negative. □

6. Small part sizes and comparison with the literature

Corollary 6.1 (Stars)**.** *Let $q \ge 1$ and $n = q+1$. Then $e_0(I_{1,q}) = 2^{q+1}$, $e_i(I_{1,q}) = 2^{q-i}\binom{q}{i}$ for $1 \le i \le q$, and $e_{q+1}(I_{1,q}) = 0$. Equivalently, for every $s \ge 1$,*

$$\ell(S/I_{1,q}^s) = 2^{q+1}\binom{s+q}{q+1} + \sum_{i=1}^{q}(-1)^i 2^{q-i}\binom{q}{i}\binom{s+q-i}{q+1-i}.$$

Proof. For $i = 1$ Theorem 1.1(b) gives $2^{n-2}\binom{n-1}{1} = 2^{q-1}q$. For $2 \le i \le q$ we have $i \le q$ and $G_{1,i} = 1$, so Lemma 5.1 gives $e_i = [z^{q-i}]\,(1-2z)^{-i-1} = 2^{q-i}\binom{q}{i}$. □

For $q = 2$ this is the case $M = \{x_1x_2, x_1x_3\}$ of [3, Theorem 3.1], with $(e_0, e_1, e_2) = (8, 4, 1)$.

Proof of Corollary 1.2. The values of e_0, e_1, e_2 and e_n are Theorem 1.1(b). Let $3 \le i \le n-1$ and $r = n-i-1$. Here $G_{2,i}(z) = 1-(i-1)(1-z) = (2-i)+(i-1)z$, so

$$\tau_{2,i}(r) = (2-i)\,2^r\binom{r+i}{i} + (i-1)\,2^{r-1}\binom{r+i-1}{i}$$

for $r \ge 1$, and $\tau_{2,i}(0) = 2-i$. If $i \le n-2 = q$, then $e_i = \tau_{2,i}(r)$ by Lemma 5.1. Using $\binom{r+i}{i} = \frac{r+i}{i}\binom{r+i-1}{i-1}$ and $\binom{r+i-1}{i} = \frac{r}{i}\binom{r+i-1}{i-1}$ we get

$$\begin{aligned}\tau_{2,i}(r) &= \frac{2^{r-1}}{i}\binom{r+i-1}{i-1}\big(2(2-i)(r+i) + (i-1)r\big)\\ &= -\frac{2^{r-1}}{i}\binom{r+i-1}{i-1}\big(2i(i-2) + (i-3)r\big),\end{aligned}$$

which is negative for $i \ge 3$ and $r \ge 1$. If $i = n-1 = q+1$, then $r = 0$ and, by Lemma 5.1, $e_i = \tau_{2,i}(0) + \tau_{q,i}(0)$, where $\tau_{q,i}(0) = G_{i-1,i}(0) = \sum_{j=0}^{i-2}(-1)^j\binom{i-1}{j} = (-1)^i$. Thus $e_{n-1} = 2-(n-1)+(-1)^{n-1} = 3-n+(-1)^{n-1} < 0$. □

Proof of Corollary 1.3. For $p = q = 2$ the only term in (1.2) is $t = 0$, $r = 1$, $\kappa = 2$, so $D_{2,2}(s) = s$, and Theorem 1.1(a) gives $\ell(S/I_{2,2}^s) = \binom{2s+3}{4} + 2s$. The coefficients are those of Corollary 1.2 with $n = 4$, and the displayed expansion is (1.1). Since $S/I_{2,2}$ is an Artinian quadratic algebra and $e_3(I_{2,2}) = -2$, the ideal $I_{2,2}$ contradicts [3, Conjecture 1.1]. □

Remark 6.2. The four-variable predictions of [3, Conjecture 3.2] that belong to the family $I_{p,q}$ are the cases $p = q = 2$ and $(p, q) = (1, 3)$. In both cases the predicted numbers of standard monomials of degree at least $2s$, namely $2s$ for $M = \{x_1x_2, x_2x_3, x_3x_4, x_4x_1\}$ and $s(2s+1) + \binom{s+1}{2}$ for $M = \{x_1x_2, x_1x_3, x_1x_4\}$, agree with $D_{p,q}(s) + D_{q,p}(s)$ in Theorem 1.1(a). For the four-cycle the listed coefficient $e_3 = 2$ should be $e_3 = -2$, as explained after Corollary 1.3. For the star $K_{1,3}$, the listed values $(e_0, e_1, e_2) = (16, 12, 6)$ agree with Corollary 6.1, which gives in addition $e_3 = 1$ and $e_4 = 0$.

The case $M = \{x_1x_2, x_1x_3, x_1x_4, x_2x_3, x_2x_4\}$, the complete graph on four vertices with the edge x_3x_4 removed, is not in the family. Its predicted length is $\binom{2s+3}{4} + s$. Under (1.1) this gives $(e_0, \dots, e_4) = (16, 12, 1, -1, 0)$, whereas the listed coefficients are $(16, 12, 1, 1)$.

Remark 6.3. Banerjee, Goel and Verma [1, Theorem 3.3, Example 3.11] computed the lengths $\ell\big(R/(\mathfrak{n}^{[Q]})^t\big)$ for the Stanley–Reisner ring $R = S/I(K_{\alpha,\beta})$ defined by the edge ideal of a complete bipartite graph, with $1 \le \alpha \le \beta$ as in [1], where $\mathfrak{n}$ is the homogeneous maximal ideal of R and $\mathfrak{n}^{[Q]}$ is generated by the Q-th powers of the variables. There the edge ideal is fixed and only $\mathfrak{n}^{[Q]}$ is raised to the t-th power, and the ring R has dimension β, so that the corresponding Hilbert–Samuel polynomial in t has degree β. In Theorem 1.1 the whole ideal $I(K_{p,q}) + (x_a^2, y_b^2)$ is raised to the s-th power in the polynomial ring S of dimension n, and the polynomial in s has degree n. For $Q = 2$ the two lengths agree when $s = t = 1$, both being $2^p + 2^q - 1$, but not in general: for $p = q = 2$ and $s = t = 2$ the formula of [1, Example 3.11] gives 23, whereas $\ell(S/I_{2,2}^2) = 39$.

Remark 6.4. The results were checked by exact computer calculations, which are not used in the proofs. Lemma 2.3 was compared with a direct recursive computation of the largest power of $I_{p,q}$ containing a given monomial, for all monomials of degree at most 9 with $p \le 4$ and $p \le q \le 5$, and Theorem 1.1(a) with a direct count of the standard monomials of $I^s_{p,q}$ for $p \le 2$, $q \le 4$ and $s \le 3$; both computations start from the generators of $I_{p,q}$ and are independent of our arguments. For $p \le 16$ and $p \le q \le 30$, the formula of Theorem 1.1(b) and the signs in Theorem 1.1(c) were compared with the coefficients of $h_{p,q}(1+u)$ computed from Corollary 3.4; this checks the derivation of Theorem 1.1(b) and (c) from Corollary 3.4.

Table 1 lists the Hilbert coefficients for $n \le 8$, computed from Corollary 3.4. The negative entries occur exactly in the rows with p even, in the positions $p{+}1, \dots, n{-}1$.

TABLE 1. The Hilbert coefficients $(e_0, \dots, e_n)$ of $I_{p,q}$ for $p + q \le 8$.

(p,q)	$(e_0, e_1, \dots, e_n)$
$(1,1)$	$(4,1,0)$
$(1,2)$	$(8,4,1,0)$
$(1,3)$	$(16,12,6,1,0)$
$(1,4)$	$(32,32,24,8,1,0)$
$(1,5)$	$(64,80,80,40,10,1,0)$
$(1,6)$	$(128,192,240,160,60,12,1,0)$
$(1,7)$	$(256,448,672,560,280,84,14,1,0)$
$(2,2)$	$(16,12,1,-2,0)$
$(2,3)$	$(32,32,6,-6,-1,0)$
$(2,4)$	$(64,80,24,-24,-17,-4,0)$
$(2,5)$	$(128,192,80,-80,-90,-32,-3,0)$
$(2,6)$	$(256,448,240,-240,-380,-204,-51,-6,0)$
$(3,3)$	$(64,80,24,1,14,6,0)$
$(3,4)$	$(128,192,80,8,33,20,2,0)$
$(3,5)$	$(256,448,240,40,130,162,78,15,0)$
$(4,4)$	$(256,448,240,40,1,-84,-82,-20,0)$

USE OF AI-ASSISTED TOOLS

AI-assisted tools were used in the research and preparation of this manuscript.

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